Maintenance Procedure — KP-40 Hydraulic Pump

Document KP-MP-012, revision A, approved by the workshop manager.

Purpose

This procedure describes the periodic maintenance of the KP-40 hydraulic pump installed on the test benches of building C.

It applies to every technician who has completed the basic hydraulics training and holds a valid workshop badge.

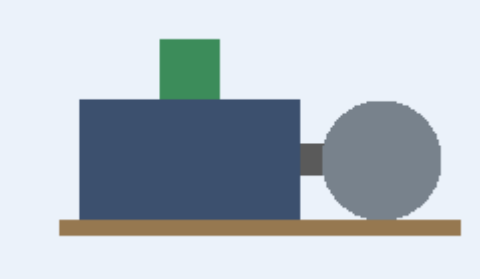

Figure 1: the KP-40 hydraulic pump.

Scope

The following operations are covered:

- Visual inspection of the pump housing and hoses
- Replacement of the suction filter
- Check of the oil level and colour
- Measurement of the outlet pressure

Repairs of the motor and of the electrical cabinet are not covered by this document; they are described in procedure KP-MP-020.

Safety

Before any operation, the pump must be stopped, isolated and locked out with a personal padlock.

The residual pressure in the circuit can reach 12 bar for several minutes after the stop[1]; wait at least ten minutes before opening any connection.

Wear safety glasses and oil-resistant gloves at all times.

Procedure

1. Stop the pump and lock out the main switch.
2. Place a drip tray under the filter housing.
3. Unscrew the filter cover by hand, without any tool.
4. Remove the old filter element and discard it with the oily waste.
5. Fit a new element, reference KP-F-110, and screw the cover back.
6. Restart the pump and check that no leak appears around the cover.

The parts needed for one maintenance are listed below.

Part	Reference	Quantity
Suction filter element	KP-F-110	1
Cover seal	KP-J-031	1
Hydraulic oil ISO VG 46	KP-H-046	2 litres

[1] Measured on the bench of building C in March 2025.

Records

Record the date, the outlet pressure and your name in the maintenance log kept next to the bench. A pressure below 180 bar **must** be reported to the workshop manager the same day.